

Starlink-Primary Resilient Gateway Reference Lab

Internship Capstone Proposal

Prepared by Havi Arji • Digi International • July 2026

Executive summary

Before I finish my internship, I want to build a **repeatable reference lab: Starlink as the primary WAN, backed by Digi 5G, managed as code through Digi Remote Manager**, with an embedded ConnectCore (Yocto) site sensor and a **measured comparison of native SureLink failover vs. Digi WAN Bonding** — validated with real fault injection, and demoed from an iPad as a **live, tappable diagram of the lab**.

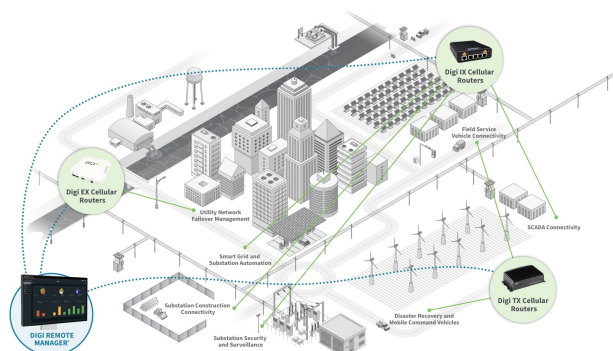
Two focused weeks. I'd like to leave Digi with a reusable asset, not just a nice internship.

Why it's worth authorizing

- **A sellable reference architecture** — “Starlink primary + Digi 5G + Opengear out-of-band” — that pre-sales and enablement can demo on demand.
- **Quantitative interoperability data** — native failover vs. WAN Bonding, by firmware, carrier, IPv4/IPv6 and MTU.
- **Concrete findings for Product/Docs** — a register of a dozen documentation gaps and contradictions I'll close against real tenants and hand back with recommended owners.

Where it fits Digi's story — and the demo you can touch

The closest of Digi's industry solution pages is **Energy & Utilities**¹ — Digi IX routers at remote sites, “Utility Network Failover Management,” Digi Remote Manager at the center, plus XBee and ConnectCore. Digi's own WAN Bonding + Starlink post² names these use cases but stops at the concept level. **This lab makes it real and measurable** — and it demos from an iPad: a live, tappable diagram where you tap a device to see its status (which WAN is carrying traffic, signal, last failover) straight from the DRM API, and watch Starlink fail over to 5G in near-real time.



Digi Energy & Utilities managed-solutions diagram — the architecture this lab embodies.

¹<https://www.digi.com/solutions/by-industry/energy-and-utilities>

²<https://www.digi.com/blog/post/digi-wan-bonding-starlink-ultra-fast-internet>

Scope: honest about two weeks

I won't pretend a 10-week program fits in two, so it's built in concentric rings:

- **Ring 0 — the 2-week commitment:** one site end to end (Starlink + 5G, DRM-as-code, BGP/IPv6, *measured* SureLink failover); the embedded ConnectCore sensor working (custom Yocto image + agent + offline buffer that loses nothing on a link cut; *secure boot deferred to Ring 2*); XBee telemetry + a Digi Container; one-command site-clone automation; a native-vs-WAN-Bonding A/B; the iPad app for site 1; and a recorded 12-minute demo.
- **Ring 1 — as hardware arrives:** sites 2–3 via the same automation, the Opendgear/Lighthouse out-of-band plane, and the app's three-site view.
- **Ring 2 — the full program:** three-site rotation, a 30-rep statistical campaign, TrustFence secure boot, secure OTA, and the full interoperability matrix.

On Opendgear: the entire critical path runs on Digi equipment, so Opendgear's lead time (Ring 1) can't block the capstone.

What I'm asking you to authorize

For the 2-week sprint, one of each is enough (more only to build sites 2–3):

- **Hardware:** 1 × IX40-05 (+ starter kit + DIN mount), 1 × ConnectCore CC-WMP255-KIT, 1 × XBee Hive + XBee 3 Zigbee kit.
- **Licenses/services:** DRM Premier + Digi Containers, a WAN Bonding Tier-3 trial + a small bonding VPS, and confirmation our tenant carries Push Monitor and ConnectCore Cloud Services entitlements.
- **Connectivity:** one Starlink Priority plan and one 5G SIM. **Already have:** Proxmox, SIEM, RADIUS.
- **In parallel (can arrive later):** Opendgear OM1304 + Lighthouse Enhance and second/third units — for Rings 1–2.

Nothing gets purchased before its verification gate closes — in particular, **no SFP+ (the IX40 is SFP-only)**.

What I need from you

A **yes** on the sprint scope and the equipment above, **sign-off** on objectives and publication rules, and **15 minutes** on day zero to align on success criteria. I already run the adjacent pieces (Lighthouse, Proxmox, SIEM, RADIUS), so the scaffolding exists. If any assumption slips (hardware lead time, a plan's IPv6 behavior), I'll adjust scope transparently rather than overclaim.

Thank you — I think this is the most useful two weeks I could spend here.